ADJOINING AN N-TH ROOT OF A UNIFORMIZER GIVES A TOTALLY RAMIFIED DEGREE N EXTENSION WHOSE INTEGRAL CLOSURE IS A[ROOT]

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'by adjoint an nth root' for 'by adjoining', and writes 'b a unit of A" where the intended reading is that u is the unit; both are reproduced unchanged. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Discrete valuation rings, uniformizers, the residue field and the ramification index. the setting at statement lines 33955-33956 and used throughout the counted proof span. Fixes the objects and the numerical invariants being computed.
- Normality of A' = A[x]/(x^n - pi) and its identification with the integral closure. used at proof lines 33978-33979. Is what upgrades the computation in A' into the structural statement about A_1.
- Kummer theory: an extension generated by an n-th root is generated by roots of a divisor when the n-th roots of unity involved lie in the base. invoked at proof lines 33985-33987. Reduces the last statement about subextensions to the unit-root claim that the congruence argument settles.
- The direct sum decomposition A' = A oplus ... oplus A(pi')^{n-1} and the shape of the image of multiplication by a power of pi'. used at proof lines 33997-34002. Is the structure that makes the congruence impossible, closing the argument.

Source reference. Original definition 33041: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33041–33041. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 0EXP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18664–18664. Cross-chapter navigation pointer into

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the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let A be a discrete valuation ring with uniformizer π. Let $n \geq 2$. Let $K_1 = K[\pi^{1/n}]$. Then*

(1) *$[K_1 : K] = n$,*
(2) *the integral closure A_1 of A in K_1 is the ring $A[\pi^{1/n}]$,*
(3) *A_1 is a discrete valuation ring,*
(4) *the ramification index of A_1 over A is n,*
(5) *K_1 is totally ramified with respect to A, and*
(6) *if n is prime to the residue characteristic of A, then K_1/K is tamely ramified and any subextension of K_1/K is generated by $\pi^{1/d}$ for some divisor d of n.*

Proof. Consider the ring $A' = A[x]/(x^n - \pi)$ and denote π' the image of x in A_1. As A' is finite free of rank n as an A-module, the element π is a nonzerodivisor in A', and hence so is π'. Moreover, $A'/\pi' A'$ is isomorphic to $A/\pi A$ which is a field. Hence A' is a discrete valuation ring with uniformizer π'. It follows that the fraction field K' of A' is an extension of degree n. Clearly, K' is obtained from K by adjoint an nth root $\pi' = \pi^{1/n}$ of π, i.e., $K' \cong K_1$. Since A' is normal, we see that A' is the integral closure of A in K', i.e., $A' = A_1$. This proves (1) – (5).

The assertion that K_1/K is tamely ramified if n is invertible in the residue field is immediate from Definition 33041 and (3) and (4). To prove the very last statement, it suffices to show that every nth root of unity $\zeta \in K'$ is contained in K, see Fields, Lemma 0EXP. It is clear that $\zeta \in (A')^*$. We can write $\zeta = a + b$ with $a \in A$ and $b \in A\pi' \oplus \ldots \oplus A(\pi')^{n-1}$. If b is nonzero, then we can write $b = u(\pi')^t$ where $t \geq 1$ and b a unit of A'. Then

$$1 = \zeta^n \equiv a^n + na^{n-1}u(\pi')^t \bmod (\pi')^{t+1}$$

Since n and a are units of A, this would imply that $b = u(\pi')^t$ is congruent to $(1 - a^n)/(na^{n-1}) \in A$ modulo $(\pi')^{t+1}$. The direct sum decomposition $A' = A \oplus \ldots \oplus A(\pi')^{n-1}$ shows that this is impossible because the image of multiplication by $(\pi')^{t+1}$ is a corresponding direct sum whose intersection with $A(\pi')^t$ is $(\pi A)(\pi')^t$. Thus $b = 0$ and $\zeta \in A$ as desired. □